

Project Overview

Project Name: OnWay - Ride Sharing Platform

Live Link : [OnWay](#)

Project Overview and Discussion

OnWay is a complete ride-sharing platform like Uber/Pathao, where riders can book rides, drivers can accept trips, support agents can assist users, and admins can manage the whole system

Deployment Guideline

- Ensure that your server is working perfectly on production and not throwing any **CORS / 404 / 504** Errors.
- Ensure that your Live Link is working perfectly and that it is not showing errors on Landing in your system.
- ⚠️ ensure that the page doesn't throw any error on reloading from any routes.
- ⚠️ Add your domain for authorization to Firebase if you use **Netlify / surge / Vercel**
- ⚠️ Logged in User must not redirect to Login on reloading any private route

Overview

OnWay is a complete ride-sharing platform inspired by Uber/Pathao, connecting passengers with drivers across Dhaka and global cities. Riders book rides instantly, drivers earn flexibly, support agents ensure safety, and admins maintain full system control.

Core Value: Safe, affordable, real-time transportation with advanced safety features (SOS, live tracking) and local+international payments (Stripe + SSLCommerz).

Role-Based Dashboard System (4 Roles)

Role	Dashboard Pages	Key Features
 Admin	5 Pages	Manage users/drivers/rides, payments, promo codes, supervise agents
 Passenger	5 Pages	Book rides, live tracking, payments, history, SOS panic button
  Driver (Rider)	10 Pages	Accept trips, OTP verification, navigation, earnings, withdrawals
 Support Agent	4 Pages	Handle complaints, live SOS, chat support, refunds, verification

Authentication System

NextAuth.js implementation for secure role-based access:

- ✓ Google OAuth (International)
- ✓ Phone OTP (Bangladesh local)

- ✓ Email/Password fallback
- ✓ Role-based redirects:
 - Passenger → /dashboard/passenger
 - Driver → /dashboard/driver
 - Admin → /admin/overview
 - Support → /support/alerts

Security Features:

- JWT tokens with role claims
- Protected routes middleware
- Session persistence (no logout on reload)
- Firebase integration for real-time auth



Complete Page Structure

Public Pages:

- 🏠 Home (Booking teaser)
- 🚗 Rider Trips & Rentals
- 📄 About/Help/Blog
- 🤖 AI Chatbot

Protected Dashboards:

- Passenger (5 pages)
- Driver (10 pages)
- Admin (5 pages)
- Support Agent (4 pages)

Next.js Tech Stack:

Frontend: Next.js 14 + Tailwind CSS

Auth: NextAuth.js

Maps: Google Maps API

Payments: SSLCommerz

Real-time: Socket.io

Database: mongodb

Deployment: Vercel, render

Page Structure

Page Structure

1. Navbar

2. Home Page

[Home Page - 1](#)

[Home Page -2](#)

3. OnWay Rider

- **Trip** (Smart Ride Booking, Fare Estimation, Real Time GPS Tracking, Payment Method)
- **Rentals**

4. Page

5. Dashboard

6. About Us

7. Help

8. Blog

8. Chat Bot (In app chat)

OnWay Website Page Structure

I. Navigation Bar

 [OnWay Logo]  Book Ride  Dashboard | About Us |  Help  EN/BN
(Optional but Important) | [Wallet: ₳500]  [User Avatar Dropdown]

II. Home Page

Description: Public landing page for all users.

Sections:

- Hero Banner (Ride booking teaser + CTA)
- Features Showcase (10 key features)
- Live Driver Map (Dhaka + Global cities)
- Rider/Passenger Signup CTAs
- Testimonials + Safety Stats
- Download App QR (iOS/Android)

Reference:

[Home Page - 1](#)

|

[Home Page - 2](#)

III. OnWay Rider Section

Description: Driver-focused public pages.

A. Trip Management

- Smart Ride Booking System
- Dynamic Fare Estimation
- Real-Time GPS Tracking
- Digital Payment System (Stripe/SSLCommerz)

B. Rentals

- Hourly/Daily vehicle rentals
- Fixed price packages
- Airport transfers
- Corporate bookings

IV. Dashboard Pages

Protected Routes (Login Required):



ADMIN (5 Pages)

- I. Dashboard Overview
- II. User Management
- III. Driver Management
- IV. Ride Management
- V. Reports & Safety

PASSENGER (5 Pages)

- |— I. Book Ride
- |— II. Active Ride
- |— III. Ride History
- |— IV. Wallet & Payments
- |— V. Profile

RIDER (10 Pages)

- |— I. Dashboard Overview
- |— II. My Rides
- |— III. Earnings
- |— IV. Withdraw
- |— V. Ride Requests
- |— VI. Schedule
- |— VII. Notifications
- |— VIII. Profile
- |— IX. Settings
- |— X. Support

SUPPORT AGENT (4 Pages)

- |— I. Live Alerts
- |— II. Case Management
- |— III. Ride Monitoring
- |— IV. Reports

V. Public Pages

A. Page

- Terms & Conditions
- Privacy Policy
- FAQ

B. About Us

- Company Mission
- Team Introduction
- Safety Features

- Coverage Areas (Dhaka + Global)

C. Help

- User Guides (Video tutorials)
- Contact Support
- Live Chat

D. Blog

- Safety Tips
- Driver Stories
- City Traffic Updates

VI. Special Features

A. Chat Bot (In-App Chat)



24/7 AI Support

- Ride booking help
- Payment issues
- SOS guidance
- Language: English/Bangla

Next.js Route Structure: (Pending) !

```
pages/
├── index.js      (Home)
├── rider/
│   ├── trip.js
│   └── rentals.js
├── dashboard/
│   ├── passenger/
│   ├── rider/
│   ├── admin/
│   └── support/
├── about.js
├── help.js
├── blog/
└── chatbot.js
```

DASHBOARD

DASHBOARD

Layout

Dashboard will be Look Like This design

Logo	Available coin userImage userRole userName	 Notification
Navigation	Sections Based on Routes	
	Footer	

Dashboard will Show Following Navigation based on user Role.

Worker	Buyer	Admin
Home	Home	Home
TaskList	Add new Tasks	Manage Users
My Submissions	My Task's	Manage Task
withdrawals	Purchase Coin	
	Payment history	

Dashboard for Admin

I. Dashboard Overview

Description: Real-time system monitoring center.

Features:

- Live metrics: Active Rides, Revenue Today, SOS Count, Users/Drivers Growth
 - Charts: Rides heatmap (Dhaka/global), Peak hours, Surge pricing trends
-

II. Passenger Management

Description: Complete passenger account control.

Features:

- Passengers list + search/filter (name/email/phone)
 - Block/Unblock + ride history view
 - Bulk actions (email verification, notifications)
-

III. Rider Management

Description: Driver onboarding and monitoring.

Features:

- Pending approvals + document verification (license/vehicle papers)
- Ratings analytics + suspension management
- Vehicle approval workflow

IV. Ride Management

Description: Full ride lifecycle control.

Features:

- All rides filter (Ongoing/Cancelled/Disputed/Completed)
 - Manual intervention + force-complete rides
 - Surge pricing override during peak hours
-

V. Reports & Safety

Description: Analytics and emergency monitoring.

Features:

- Revenue/Commission reports (CSV/PDF export)
- SOS dashboard + resolution statistics
- Compliance audit logs + safety trends

Dashboard for Rider

I. Dashboard (Overview)

Description: Main home page after login.

Features:

- Online/Offline Toggle Button
- Today's Earnings Summary
- Total Earnings

- Total Completed Rides
 - Average Rating
 - Recent Rides List
 - Ongoing Ride Status (if available)
-

II. My Rides

Description: View complete ride history.

Sections:

- Ongoing Rides
- Completed Rides
- Cancelled Rides

Ride Information:

- Passenger Name - Pickup/Drop Location - Distance - Fare - Date & Time - Ride Status
-

III. Earnings

Description: Detailed earning analytics.

Features:

- Daily/Weekly/Monthly Earnings
 - Earnings Chart
 - Commission Breakdown
 - Bonus Earnings
 - Available Balance
-

IV. Withdraw

Description: Request payouts from earnings.

Features:

- Available Balance Display
- Withdraw Request Form
- Payment Method Selection (bKash/Stripe)

- Withdraw History + Invoices
 - Withdraw Status Tracking
-

V. Ride Requests

Description: Incoming ride notifications.

Features:

- Ride Request Cards
 - Pickup → Drop Location Map
 - Distance + Estimated Fare
 - Accept/Reject Buttons
 - Countdown Timer
-

VI. Schedule

Description: Manage working availability.

Features:

- Select Working Days
 - Set Working Hours (e.g., 8AM-10PM)
 - Vacation Mode Toggle
 - Availability Status Indicator
-

VII. Notifications

Description: All platform notifications.

Features:

- Ride Updates (New request/Completed)
 - Payment Notifications
 - Bonus Announcements
 - Admin Messages
 - Read/Unread Indicators
-

VIII. Profile

Description: Personal & vehicle information.

Features:

- Full Name + Phone Number
 - Driving License Details
 - Vehicle Information (Car/Bike)
 - Document Upload (License/Registration)
 - Profile Picture
 - Verification Status
-

IX. Settings

Description: Account preferences.

Features:

- Change Password
 - Notification Preferences
 - Language Selection (English/Bangla)
 - Dark Mode Toggle
-

X. Support

Description: Platform support access.

Features:

- Create Support Ticket
- View Ticket History
- Ticket Status Updates
- FAQ Section

Dashboard for Passenger

I. Book Ride

Description: Quick and intuitive ride booking.

Features:

- Map picker + real-time fare estimator
 - Vehicle type selection + schedule option (now/later)
 - Promo code input + payment method select
-

II. Active Ride

Description: Live ride tracking and communication.

Features:

- Live GPS tracking + accurate ETA updates
 - Driver info + in-app chat/call functionality
 - Panic button + share trip with contacts
-

III. Ride History

Description: Complete ride records and feedback.

Features:

- Past rides list + digital receipts download
 - Rate driver + add detailed feedback
 - Dispute ride option with support ticket
-

IV. Wallet & Payments

Description: Payment management and credits.

Features:

- Add Stripe/SSLCommerz payment methods
 - Ride credits balance + promo code history
 - Transaction statements and refund status
-

V. Profile

Description: Personal preferences and settings.

Features:

- Saved addresses + emergency contacts
- Ride preferences (AC/non-AC, quiet driver)
- Notification settings + language selection

Dashboard for Support Agent

I. Live Alerts

Description: Real-time emergency and complaint monitoring.

Features:

- SOS + complaints real-time feed with sound alerts
 - Live location tracking on interactive map
 - One-click call driver/passenger via in-app VoIP
-

II. Case Management

Description: Handle and resolve support tickets.

Features:

- Open tickets + priority queue (High/Medium/Low)
 - Direct chat with involved parties (driver/passenger)
 - Resolution notes + close case with feedback
-

III. Ride Monitoring

Description: Supervise ongoing platform activity.

Features:

- Ongoing rides overview with status filters
 - Low-rating driver alerts + performance warnings
 - Manual dispatch option for high-demand areas
-

IV. Reports

Description: Safety and performance analytics.

Features:

- Daily SOS summary + response time metrics
- Resolution time analytics + agent performance
- Escalation log to Admin with critical incidents

20 Key Features

20 Key Features

1. Smart Ride Booking (Instant Request)

Feature:

Users can request a ride instantly through the mobile app.

Benefits:

- Saves time
 - No need to search for vehicles manually
-

2. Real-Time GPS Tracking

Feature:

Live tracking of the driver's location on the map.

Benefits:

- Easy to locate the driver
 - Reduces uncertainty and waiting confusion
-

3. Fare Estimation

Feature:

Estimated fare is shown before confirming the ride.

4. In-App Call and Chat

Feature:

Users and drivers can communicate within the app without sharing phone numbers.

Benefits:

- Maintains privacy
 - Improves coordination
-

5. Digital Payment System (SSLcommerz)**Feature:**

Supports cash, card, and mobile wallet payments.

Benefits:

- Fast and secure transactions
 - Reduces cash handling issues
-

6. Ride History and Invoice**Feature:**

Complete record of previous rides with receipts.

Benefits:

- Easy expense tracking
 - Useful for business and frequent travelers
-

7. Safety Features (SOS)**Feature:**

Emergency button and live trip sharing with contacts.

Benefits:

- Increases user safety
- Builds confidence, especially for female riders

8. Real-Time Driver Matching

- The system matches nearby drivers and OTP ensures correct trip start.

9. Face Verification (with rider)

10. Driving License Extraction (using any AI Model or any parsing services or npm packages)

11. Trafficking on road

12. KYC Verification (with rider)

13 . Item Recovery

14. OTP Verification

15. Driver Rating and Review

15. Promo Code & Discounts

17. Chatbot (Ai Supported)

18. Newsletter

19. Demand prediction

20. Dynamic pricing based on the weather

About Project

OnWay A Ride Sharing Platform

Core Contributions :

1. Secure Payment Gateway Integration

Workflow: To ensure a frictionless checkout experience, we integrated **SSLCommerz**, the leading payment gateway in Bangladesh. Upon ride completion, the system calculates the final fare and redirects the passenger to a secure payment hosted page. The platform supports **MFS (bKash, Nagad), Debit/Credit Cards, and Cash-on-Delivery**.

Technical Implementation: The backend handles payment initiation via a **POST /api/payment/initiate** route, which communicates with the SSLCommerz Sandbox/Live API. We used **IPN (Instant Payment Notification)** to verify transaction authenticity before updating the ride status in MongoDB.

Tech Stack: SSLCommerz SDK, Axios, Express.js, MongoDB.

2. SOS Emergency Response System

Workflow: Safety is a core pillar of OnWay. We developed a real-time SOS feature accessible to both riders and drivers during an active trip. When triggered, the system bypasses standard support queues and sends an immediate high-priority alert to the **Support Agent Dashboard**.

Technical Implementation: This is powered by **Socket.io** for bi-directional, real-time communication. The alert broadcasts a **sos:alert** event containing the user's live GPS coordinates and trip ID. Support agents receive a visual "Red Alert" notification and can track the user's movement instantly on a live map.

Tech Stack: Socket.io (WebSockets), Leaflet.js (Live Tracking), [Node.js](#).

3. Passenger Refund Management System

Workflow: To handle disputes (e.g., driver no-show or overcharging), we built a structured Refund Module. Passengers can submit requests from their dashboard by selecting the ride type and reason. These requests appear in the **Support Dashboard** with a "Pending" status. Agents can review, add internal notes, and either **Approve** or **Reject** the claim.

Technical Implementation: We used a **State Machine** logic in MongoDB to manage refund statuses (**pending**, **approved**, **rejected**). The backend uses **PATCH** requests to update the refund state, and the passenger's UI reflects these changes in real-time using React state management.

Tech Stack: MongoDB, React.js, [Express.js](#).

4. Real-Time Road Traffic Visualization

Workflow: To help users plan their journeys better, we implemented a dual-layer traffic visualization system. Users can toggle a **Live Traffic** button to see current road congestion. The map displays color-coded lines: **Green** (Clear), **Yellow** (Moderate), and **Red** (Heavy Jam).

Technical Implementation: We used a hybrid approach. For the visual overlay, we utilized **Google Traffic Tiles** integrated into **Leaflet.js**. For data accuracy, we integrated the **TomTom Flow API**, which calculates the ratio between **currentSpeed** and **freeFlowSpeed**. If the ratio falls below a certain threshold, the polyline color on the map updates dynamically to alert the user.

Tech Stack: Leaflet.js, TomTom Traffic API, Google Tile Services.

Technical Challenges & Solutions :

1. Payment Gateway Integration

Challenge: Ensuring transaction security and handling "Incomplete Payments." Sometimes users close the browser after paying but before the success redirect.

Solution: We implemented **IPN (Instant Payment Notification)**. Even if the user closes the window, SSLCommerz sends a background server-to-server notification to our API to confirm the payment and update the database.

2. SOS Emergency Service

Challenge: Maintaining a constant connection in low-network areas. If the passenger's internet is unstable, the SOS alert might not reach the Support Agent instantly.

Socket Reliability: We used **Socket.io's Acknowledgement** and **Reconnection logic**. If the connection drops, the system attempts to reconnect automatically and caches the alert to send it as soon as the heartbeat is restored.

3. Refund Management System

Challenge: Preventing "Fraudulent Refund Requests." Passengers might try to claim refunds multiple times for the same ride.

Logic Solution: We implemented a **Strict Validation Layer**. A refund request can only be submitted if the ride status is "Completed" or "Cancelled by Driver," and our database schema ensures a **One-to-One relationship**—meaning only one refund entry can exist for a unique `Booking_ID`.

4. Real-time Traffic Visualization

Challenge: Performance lag and API Rate Limiting. Rendering high-resolution traffic tiles and calculating speed ratios for every route can slow down the map's performance on mobile devices.

Optimization: We optimized this by using **Layer Toggling**. The traffic tiles are only fetched and rendered when the user explicitly clicks the "Live Traffic" button. Additionally, we used **Debouncing** on the TomTom API calls to ensure we don't hit rate limits during map panning or zooming.

General Implementation Challenges :

CORS & Environment Sync: Syncing the backend API with third-party services (SSLCommerz/TomTom) across different environments (Local vs. Production) required rigorous **Environment Variable** management and CORS policy configuration.

Cross-Dashboard Communication: Ensuring that an action in the **Rider Dashboard** (like an SOS or Refund) triggers an instant visual update in the **Support Agent Dashboard** without manual refreshing. We solved this using a central **Socket.io Room** architecture.

5. In-App Chat & Call System

Built on Socket.io with each active ride isolated in its own WebSocket room — no cross-ride messaging possible. I implemented phone number masking via a server-side relay so neither riders nor drivers ever see each other's real contact. Passengers can share location photos through Cloudinary integration directly in chat.

Custom hooks built: **useChat** (WebSocket lifecycle management), **useRequireRole** (role-based room access). Custom component: **CallModal** (in-app call UI with mute and end-call controls). Message delivery optimized to under 100ms.

Tech: Socket.io, Cloudinary, Next.js 14, lucide-react, react-hot-toast

Scenario: Driver can't find passenger. Passenger sends a location photo via chat and calls via in-app relay. Zero personal information shared.

6. AI Support Chatbot (GPT-4o)

A 24/7 AI assistant powered by OpenAI GPT-4o. A strict system prompt constrains it to OnWay-specific topics only. Full conversation history is passed per API call for context-aware multi-turn dialogue. Language is auto-detected — responses in Bangla or English based on the user's message. Responses are streamed for real-time typing feel. Handles booking help,

payment issues, promo codes, SOS guidance, and account recovery. Escalates unresolvable issues to a human agent with full conversation context attached.

Tech: OpenAI GPT-4o API, Next.js API Routes, Streaming Response

Scenario: User writes in Bangla about a failed payment. Chatbot responds in Bangla, provides troubleshooting steps, and escalates to human support if unresolved.

7. OTP Verification System

Required for all new registrations, password resets, wallet withdrawals, and sensitive driver actions. 6-digit OTPs delivered via international SMS Gateway (Twilio/equivalent), expiring after 5 minutes via MongoDB TTL index. Rate limited to 3 requests per 10-minute window — prevents SMS bombing. OTPs are stored hashed (bcryptjs), never plaintext. Three failed attempts auto-invalidate the OTP — prevents brute-force attacks.

Tech: SMS Gateway, MongoDB TTL Index, NextAuth.js, bcryptjs, Next.js API Routes

8. Promo Code & Discount Engine

I designed a flexible MongoDB schema supporting all campaign types: percentage discounts, flat discounts, first-ride-only restrictions, per-user usage limits, total usage caps, and expiry dates with automatic deactivation. Server-side validation checks all conditions in sequence before applying any discount — client cannot bypass. Discount is shown transparently in the fare breakdown before ride confirmation.

Tech: MongoDB/Mongoose, Next.js API Routes, Framer Motion, react-hot-toast

Scenario: Code "RAIN30" — 30% off, 500 max uses, one per user. Full server-side validation on apply. usedByUsers array updates atomically on confirmation.

Tech Stack (My Modules)

Next.js 14 — SSR, API routes, RBAC middleware. Socket.io — Chat, call relay, live events. OpenAI GPT-4o — Bilingual AI chatbot. NextAuth.js + JWT — Role-based sessions. MongoDB + Mongoose — All data schemas. Tailwind CSS + Framer Motion — UI and animations. Cloudinary — Chat image sharing. Google Maps API — Distance, directions, map picker. Weather API — Surge pricing triggers. SMS Gateway — OTP delivery. Custom React Hooks — useChat, useAuth, useFareEstimate.

Key Challenges & Resolutions

Socket.io Scalability: Thousands of connections caused memory pressure. Fixed by isolating each ride into its own room and optimizing event listeners. Redis adapter planned for horizontal scaling.

Cross-Role Security: Passenger sessions could theoretically reach Driver routes. Fixed with a custom Next.js Middleware that validates JWT role claim before any page hydrates.

OTP Reliability in Bangladesh: Local carriers dropped some SMS. Fixed with retry logic, fallback carrier, and whitelisted sender ID.

Chatbot Scope Creep: GPT-4o answered off-topic questions, wasting tokens. Fixed with a strict system prompt limiting responses to OnWay-only topics.

Project Entry (Resume Format)

OnWay — Ride Sharing Platform | Feb 15, 2026 – Present [Live Link](#) | [GitHub Link](#)

Overview: A collaborative full-stack ride-sharing platform by Team DevVibe, featuring AI-driven automation and real-time coordination comparable to Uber/Pathao for the Bangladeshi market.

My Core Contributions:

AI & Pricing: Built a dynamic pricing engine adjusting fares in real-time based on Weather API data and live traffic density. Implemented a demand prediction model identifying high-demand zones to help drivers position efficiently.

Communication: Built a secure in-app chat and call system (Socket.io + phone number masking) and integrated an AI Chatbot (GPT-4o) for 24/7 bilingual support in English and Bangla.

Safety & Growth: Implemented OTP Verification via SMS Gateway, a weighted Driver Rating system, a flexible Promo Code engine, and an automated Newsletter subscription system.

Collaborative Team Features: Real-time GPS tracking, SSLCommerz payments, AI Face Verification, SOS emergency support, Item Recovery module.

Tech Stack: Next.js 14, Node.js, Socket.io, NextAuth.js, MongoDB, OpenAI GPT-4o, Weather API, SMS Gateway, Cloudinary, Framer Motion, Tailwind CSS

Conclusion

My work on OnWay directly impacts platform profitability (Dynamic Pricing), user safety (OTP + Private Communication), service quality (Driver Rating), and long-term growth (Promo + Newsletter). By integrating AI intelligence with real-time infrastructure and a strong security foundation, I helped build a mobility solution that is not just functional but intelligent, scalable, and market-ready for Bangladesh's competitive ride-sharing industry.